MATH MA'AM ACADEMY

Where Every Subject Clicks.

STD 5 MATHEMATICS – CHAPTER 3: FRACTIONS AND DECIMALS

Learning objectives: Fractions • Types of Fractions • Comparison of Fractions • Operations on Fractions • Decimals • Conversion of Decimals • Types of Decimals • Comparing Decimals • Addition and Subtraction of Decimals

3.1 FRACTION

A **fraction** represents a part of a whole. The whole is divided into equal parts.

Fraction = Number of equal parts taken / Total number of equal parts

In a fraction a/b, **a** is the numerator (parts taken) and **b** is the denominator (total equal parts). The denominator cannot be zero.

Example: If a pizza is divided into 8 equal slices and 3 slices are eaten, the fraction eaten is 3/8.

Remember: Fractions describe equal parts. If the parts are not equal, the simple fraction model does not apply directly.

3.2 TYPES OF FRACTIONS

Type	Meaning	Example
Proper fraction	Numerator is smaller than denominator.	3/7, 5/9
Improper fraction	Numerator is greater than or equal to denominator.	7/5, 8/3, 6/6
Like fractions	Fractions with the same denominator.	5/7, 4/7
Unlike fractions	Fractions with different denominators.	4/7, 8/11
Unit fraction	Numerator is 1.	1/4, 1/9
Equivalent fractions	Fractions with the same value.	5/7 = 15/21
Mixed fraction	A whole number and a proper fraction together.	4 3/4, 6 1/9

Converting mixed fractions and improper fractions

Mixed to improper: Multiply the whole number by the denominator, add the numerator, and keep the same denominator.
Example: 3 2/5 = (3 × 5 + 2)/5 = 17/5.

Improper to mixed: Divide the numerator by the denominator. The quotient is the whole number and the remainder is the new numerator.
Example: 17/4 = 4 1/4.

Equivalent fractions

Multiply or divide both the numerator and denominator by the same non-zero number to obtain an equivalent fraction.
Example: 4/7 = (4 × 3)/(7 × 3) = 12/21.

3.3 COMPARISON OF FRACTIONS

Like fractions

When denominators are the same, compare numerators. The fraction with the larger numerator is greater.
Example: 5/7 < 6/7.

Same numerator

When numerators are the same and denominators are positive, the fraction with the smaller denominator is greater.
Example: 4/5 > 4/7.

Unlike fractions – cross multiplication

To compare a/b and c/d, compare a × d with c × b (for positive denominators).
Example: Compare 4/5 and 5/7: 4 × 7 = 28 and 5 × 5 = 25. Since 28 > 25, 4/5 > 5/7.

Unlike fractions – making like fractions

Find the L.C.M. of the denominators and convert both fractions to equivalent fractions with that denominator. Then compare their numerators.
Example: 4/5 = 28/35 and 5/7 = 25/35. Therefore, 4/5 > 5/7.

Ordering fractions: Convert fractions to a common denominator (often the L.C.M. of denominators), then arrange the numerators in ascending or descending order.

3.4 OPERATIONS ON FRACTIONS

Addition and subtraction of like fractions

Keep the denominator unchanged and add or subtract the numerators. Simplify the answer if possible.
Example: 7/15 + 4/15 = 11/15
Example: 12/17 – 9/17 = 3/17.

Addition and subtraction of unlike fractions

First convert the fractions into like fractions using a common denominator (usually the L.C.M.). Then add or subtract the numerators.
Example: 2/3 + 1/4 = 8/12 + 3/12 = 11/12.
Example: 4/5 – 5/7 = 28/35 – 25/35 = 3/35.

Operations on mixed fractions

Method 1: Add or subtract whole-number parts and fractional parts separately. Convert the fractional parts to like fractions, then simplify or regroup if needed.
Example: 3 1/5 + 4 1/6 = 7 + (6/30 + 5/30) = 7 11/30.

Method 2: Convert mixed fractions into improper fractions, find a common denominator, and then add or subtract.
Example: 3 1/5 = 16/5 and 4 1/6 = 25/6. Sum = 96/30 + 125/30 = 221/30 = 7 11/30.

Multiplication of fractions

Multiply the numerators together and multiply the denominators together. Simplify the result.
Example: 2/5 × 4/7 = (2 × 4)/(5 × 7) = 8/35.

Fraction × whole number: Multiply the numerator by the whole number and keep the denominator (or write the whole number over 1).
Example: 15 × 3/17 = 45/17 = 2 11/17.

Division of fractions

To divide by a fraction, multiply by its reciprocal. The reciprocal is obtained by interchanging numerator and denominator.

Whole number ÷ fraction: 18 ÷ 2/9 = 18 × 9/2 = 81.
Fraction ÷ whole number: 3/14 ÷ 6 = 3/14 × 1/6 = 1/28.
Fraction ÷ fraction: 5/36 ÷ 7/18 = 5/36 × 18/7 = 5/14.

Important: In fraction division, keep the first fraction, change ÷ to ×, and invert only the second fraction. Never divide by zero.

3.5 DECIMALS

A **decimal** is another way to represent a number. It has a whole-number part and a fractional part separated by a decimal point.

Example: 15.25 is read as “fifteen point two five”. The whole-number part is 15 and the decimal part is 0.25.

Place	Fraction	Decimal
Tenths	1/10	0.1
Hundredths	1/100	0.01
Thousandths	1/1000	0.001

Example: 3/10 = 0.3; 13/1000 = 0.013. In 4.275, 2 is in the tenths place, 7 in the hundredths place and 5 in the thousandths place.

3.6 CONVERSION OF DECIMALS

Whole numbers into decimals

A whole number can be written with a decimal point followed by zero(s) without changing its value.
Example: 18 = 18.0 = 18.00.

Fractions into decimals

For denominators 10, 100 or 1,000, write the numerator with the correct number of digits after the decimal point.
Examples: 7/10 = 0.7; 25/100 = 0.25; 13/1000 = 0.013.

For other denominators, divide the numerator by the denominator, or make an equivalent fraction with denominator 10, 100, 1,000, etc., when possible.

Decimals into fractions

Write the decimal number over 1, then multiply numerator and denominator by 10, 100 or 1,000 according to the number of digits after the decimal point. Reduce to lowest terms.
Example: 25.013 = 25013/1000.
Example: 0.45 = 45/100 = 9/20.

Shortcut: When converting a fraction with denominator 10, 100 or 1,000 into a decimal, place the decimal point 1, 2 or 3 places from the right respectively.

3.7 TYPES OF DECIMALS

Type	Meaning	Example
Equivalent decimals	Different-looking decimals with the same value.	0.8 = 0.80 = 0.800
Like decimals	Decimals having the same number of digits after the decimal point.	0.714 and 0.568
Unlike decimals	Decimals having different numbers of digits after the decimal point.	0.8 and 0.24

Making like decimals: Add zeros to the right of the decimal part. This does not change the value.
Example: 2.5, 0.25 and 3.125 can be written as 2.500, 0.250 and 3.125.

3.8 COMPARING DECIMALS

To compare decimal numbers:

• Compare the whole-number parts first. The number with the greater whole-number part is greater.

• If whole-number parts are equal, compare tenths, then hundredths, then thousandths from left to right.

• Add zeros to the right when needed to make the decimal places equal.

Example: Compare 56.568 and 56.548. Whole parts and tenths are equal. At the hundredths place, 6 > 4. Therefore, 56.568 > 56.548.

Example: 3.75 = 3.750, because adding a zero to the right of the decimal does not change the value.

3.9 ADDITION AND SUBTRACTION OF DECIMALS

Follow these steps:

• Make the decimals like decimals by adding zeros to the right if necessary.

• Write the numbers one below the other, lining up the decimal points.

• Add or subtract as with whole numbers, carrying or borrowing where required.

• Place the decimal point in the answer directly below the decimal points in the numbers.

Addition example:
12.35
+ 8.70
= 21.05

Subtraction example:
814.49
– 42.35
= 772.14

Common mistake: Do not align numbers by their last digit. Always align decimal points before adding or subtracting.

QUICK REVISION – IMPORTANT RULES

Topic	Rule to remember
Fraction	Numerator = parts taken; denominator = total equal parts.
Mixed to improper	(Whole number × denominator + numerator) / denominator
Like fraction addition/subtraction	Add/subtract numerators; keep denominator.
Unlike fraction addition/subtraction	Make denominators same first.
Fraction multiplication	Numerator × numerator; denominator × denominator.
Fraction division	Keep, change, flip: multiply by reciprocal of divisor.
Decimal places	Tenths = 1 digit; hundredths = 2; thousandths = 3.
Equivalent decimals	Adding zeros to the right does not change value.
Decimal comparison	Compare from left to right; use equal decimal places.
Decimal addition/subtraction	Line up decimal points.

IMO preparation checklist: Practise identifying shaded fractions, converting mixed and improper fractions, finding equivalent fractions, comparing unlike fractions, all four fraction operations, decimal place values, fraction–decimal conversions, equivalent/like/unlike decimals, decimal comparison, and decimal addition and subtraction.

ALL THE BEST